Disorder in a Spinel Quantum Magnet & Single Crystal Growth of a Topological Magnon Candidate

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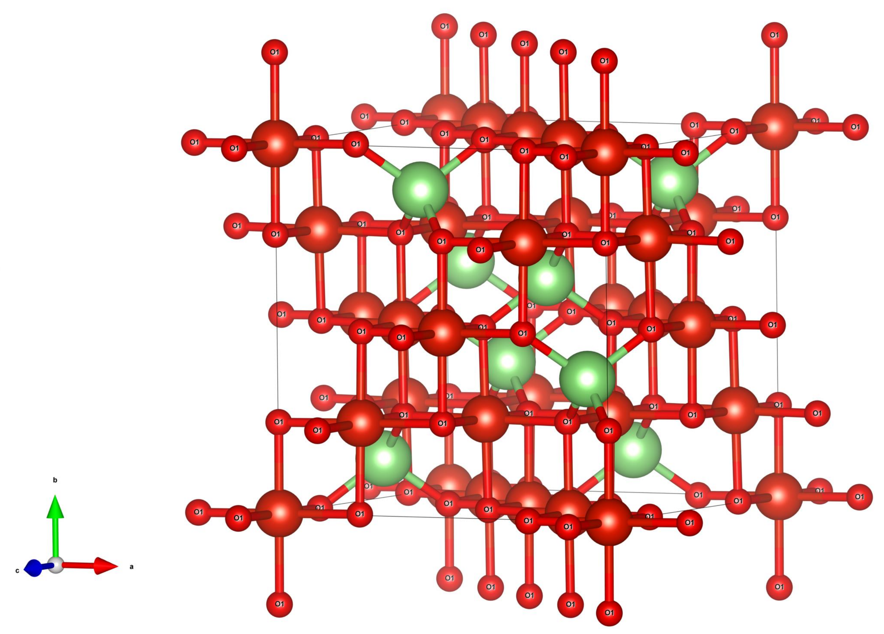
Introduction: Spinel compounds & Magnons

- Spinel compounds are materials composed of an oxide ion and a metal cation with a cubic crystal structure.
- The general formula is AB_2O_4, where A lies in tetrahedral sites and B lies in octahedral sites.
- We want to investigate the effects of disorder of the $ZnAl_2O_4$ spinel when doped with Cr.
- Topological magnons are quasiparticle excitations in magnetic materials with non-trivial band structure.
- We grow a single crystal of the topological magnon candidate $NdFeO_3$.

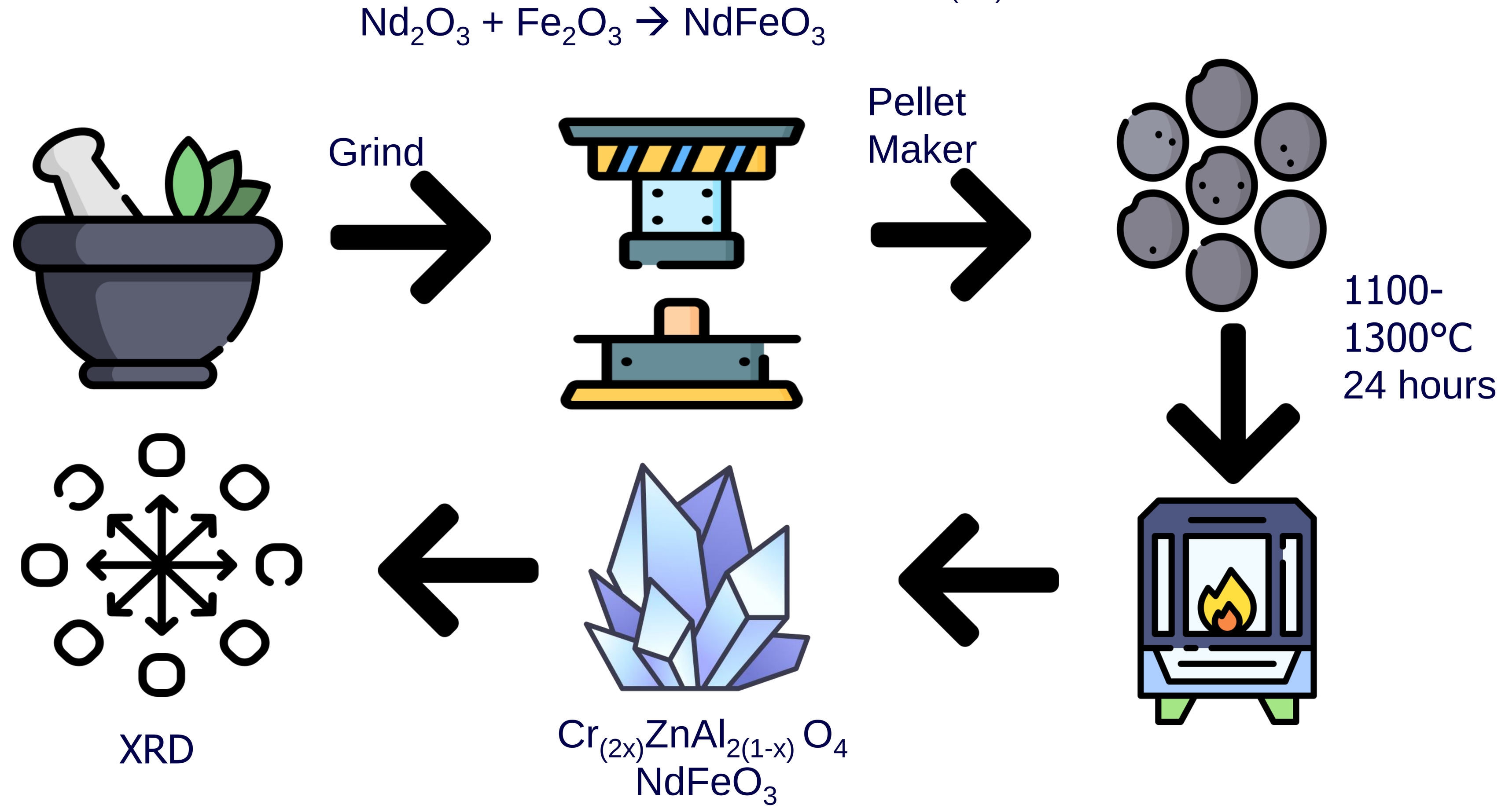


$ZnAl_2O_4$ Crystal Structure

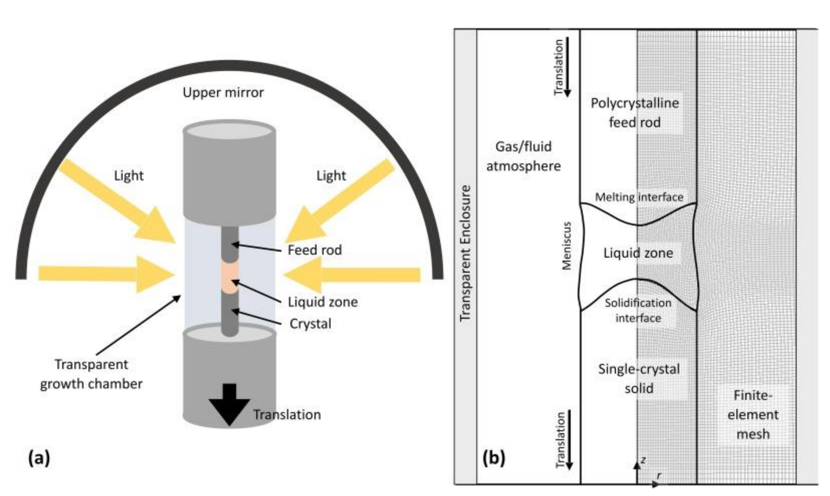
Methods

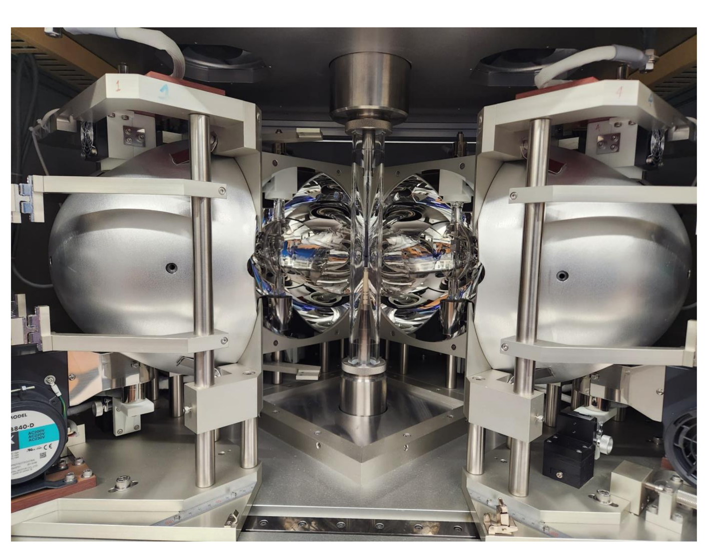
Solid State Synthesis and Diffraction

$ZnO + Al_2O_3 + Cr_2O_3 \rightarrow ZnCr_{2x}Al_{2(1-x)}O_4$
$Nd_2O_3 + Fe_2O_3 \rightarrow NdFeO_3$

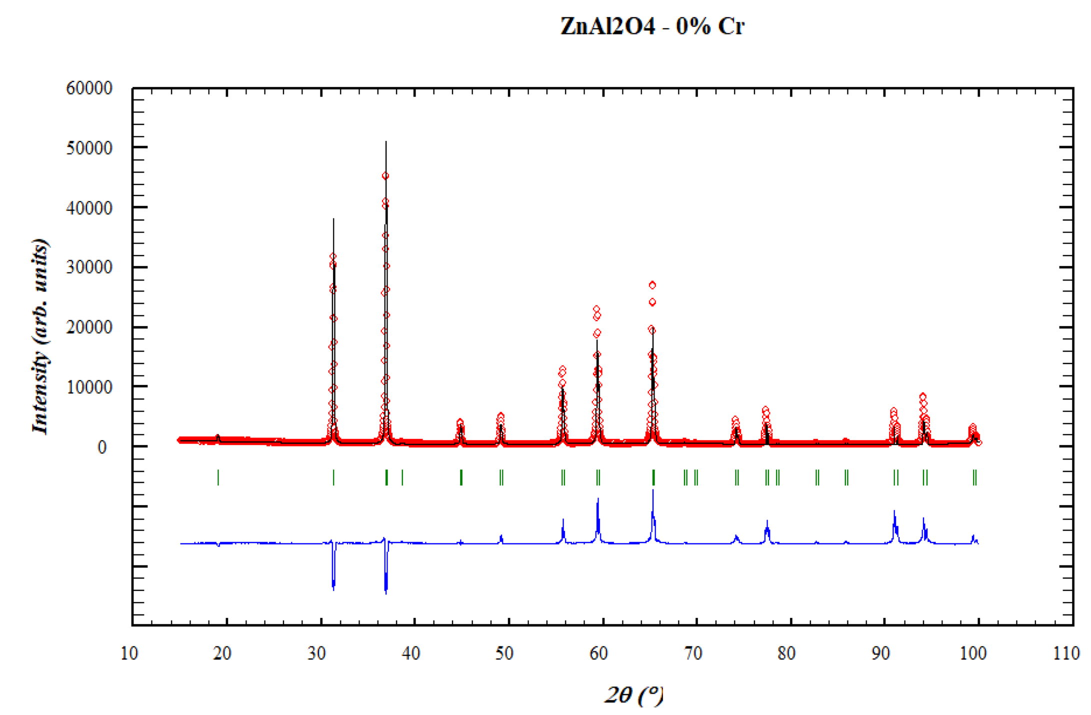


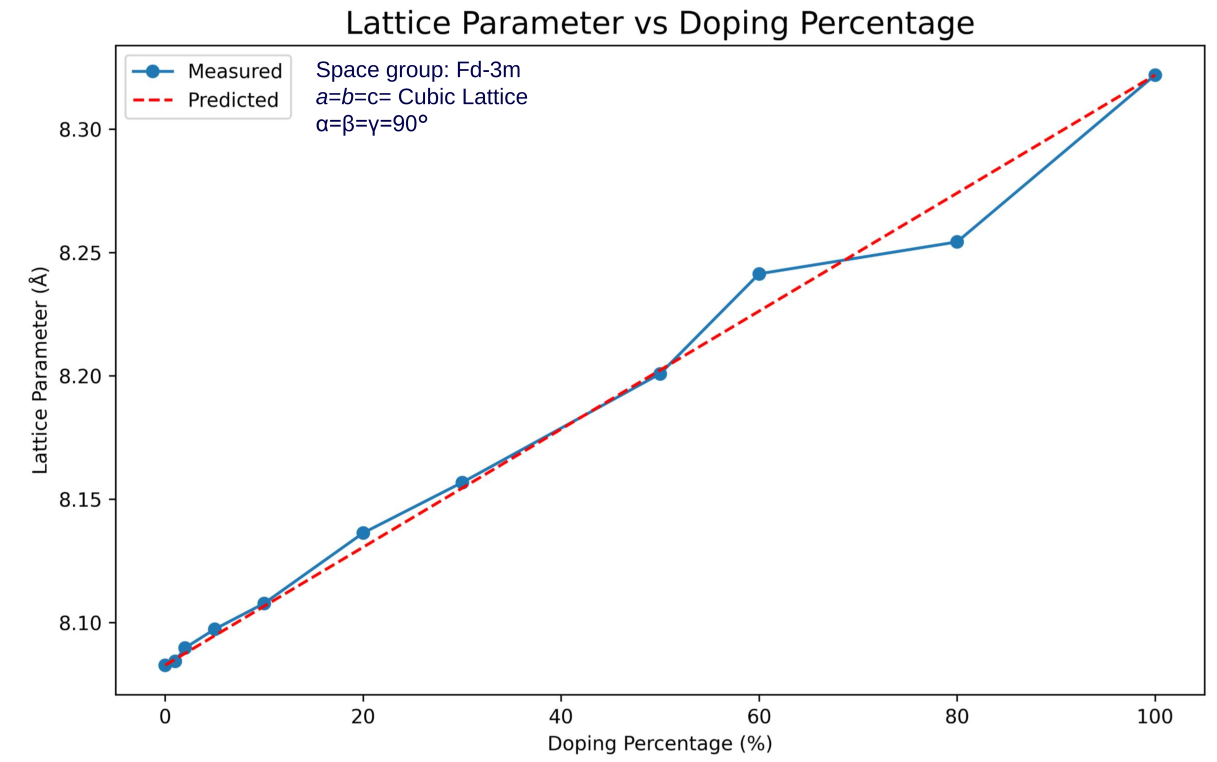
Optical Floating Zone Furnace



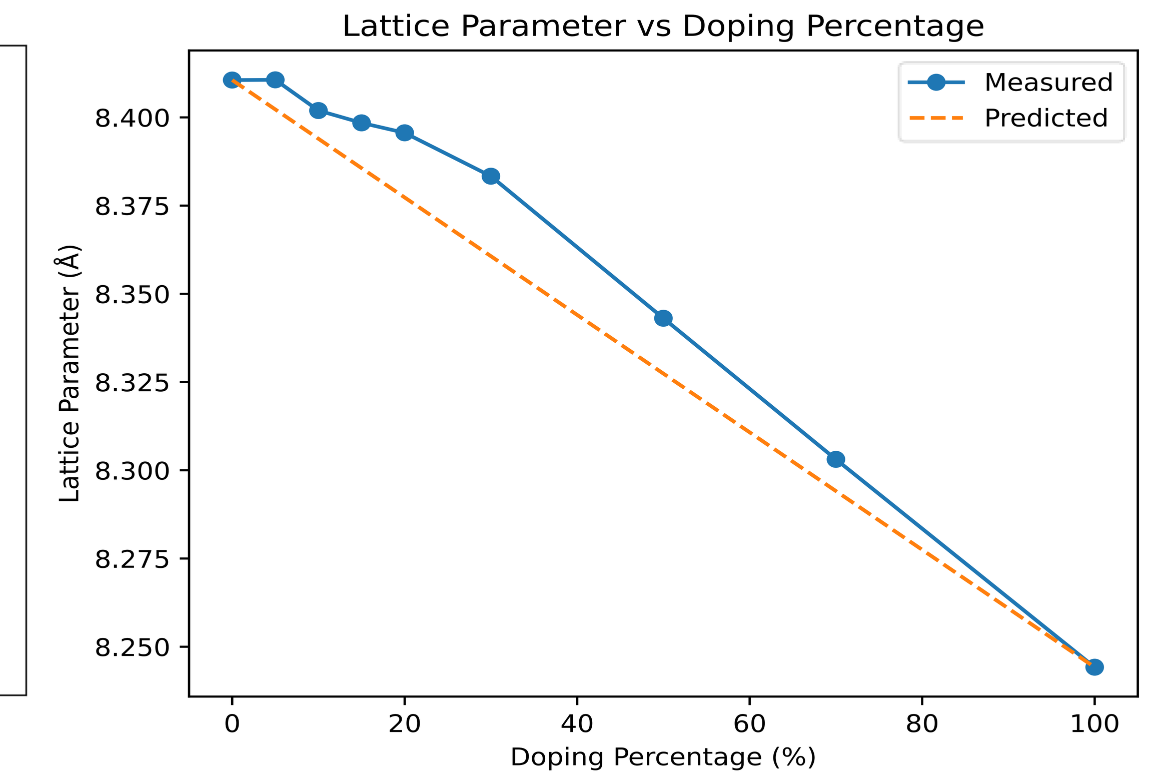


Results, Analysis, and Discussions

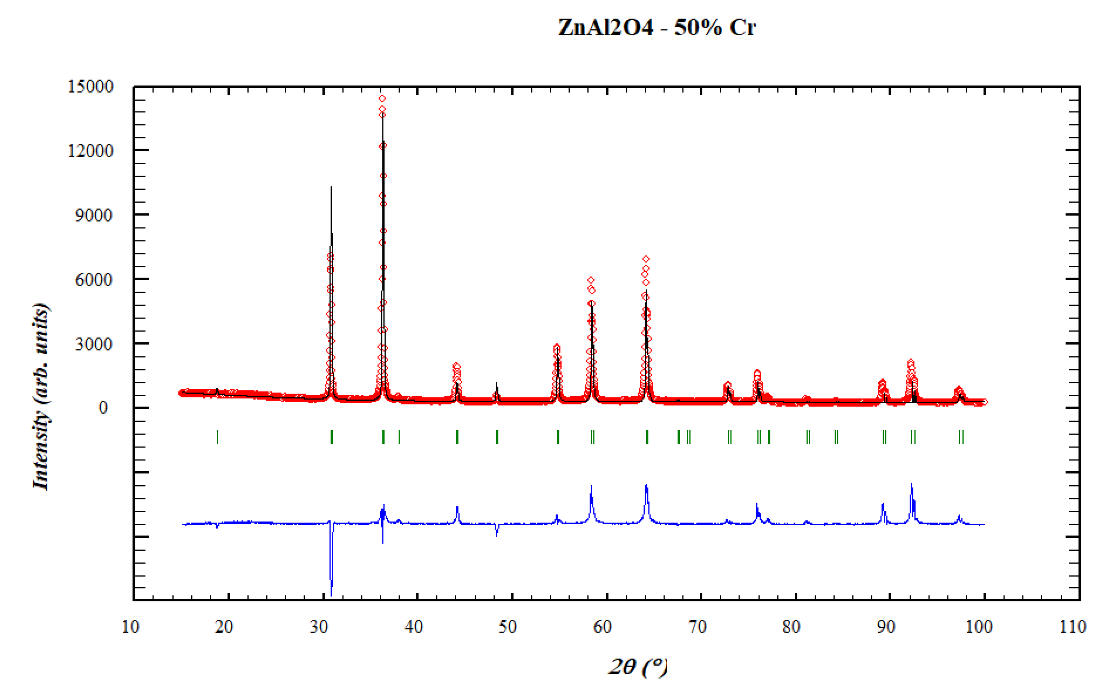


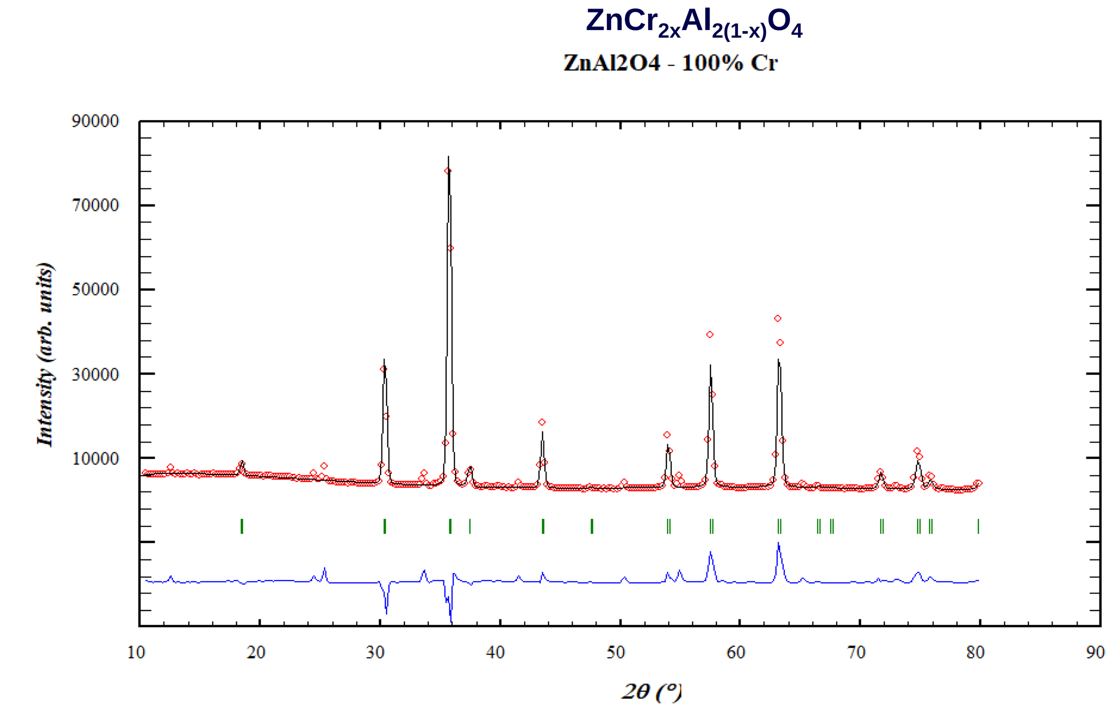


$ZnCr_{2x}Al_{2(1-x)}O_4$



$Li_xZn_{(1-x)}V_2O_5$





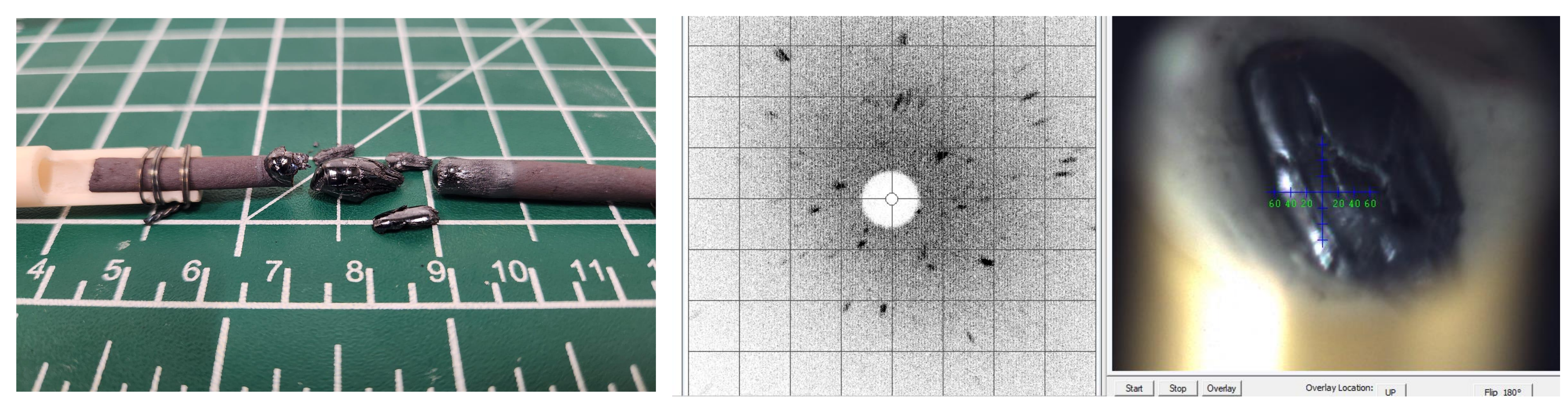


Figure: Single Crystal Growth Rod of $NdFeO_3$ and its preliminary Laue Scattering

Analysis and Discussions

The structural characterization were carried out using X-ray diffraction analysis.

A linearly increasing trend of lattice constant was observed with the increase in Cr-doping to $ZnAl_2O_4$.

It appears to follow Vegards's rule with respect to changes in lattice parameter on doping.

- The average lattice deviation was measured to be 0.00466 **Å**.
- The increasing trend can be attributed to the larger atomic radius of Cr as compared to Al.
- The deviation might be attributed to the difference in atomic radii between Cr and Al, leading to lattice strain and distortion, as well as Cr being anti-ferromagnetic and Al being paramagnetic at room temperature.
- The single crystal did turn out to be showing distinct patterns. We will replicate the method to make a longer and smoother crystal for magnetometry.

Conclusions

1. Cr-doped $ZnAl_2O_4$ shows a linear increase in lattice constant, following Vegard's Law.
2. The comparison from Li-doped ZnV_2O_5 indicates something interesting happening at low doping percentages.
3. Further study is required to understand lattice distortion through thermodynamic magnetometric studies.
4. Future work will also involve re-growth of NdFeO3 crystal and magnetometric measurements for verification of the predicted topological magnon nature.

References

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